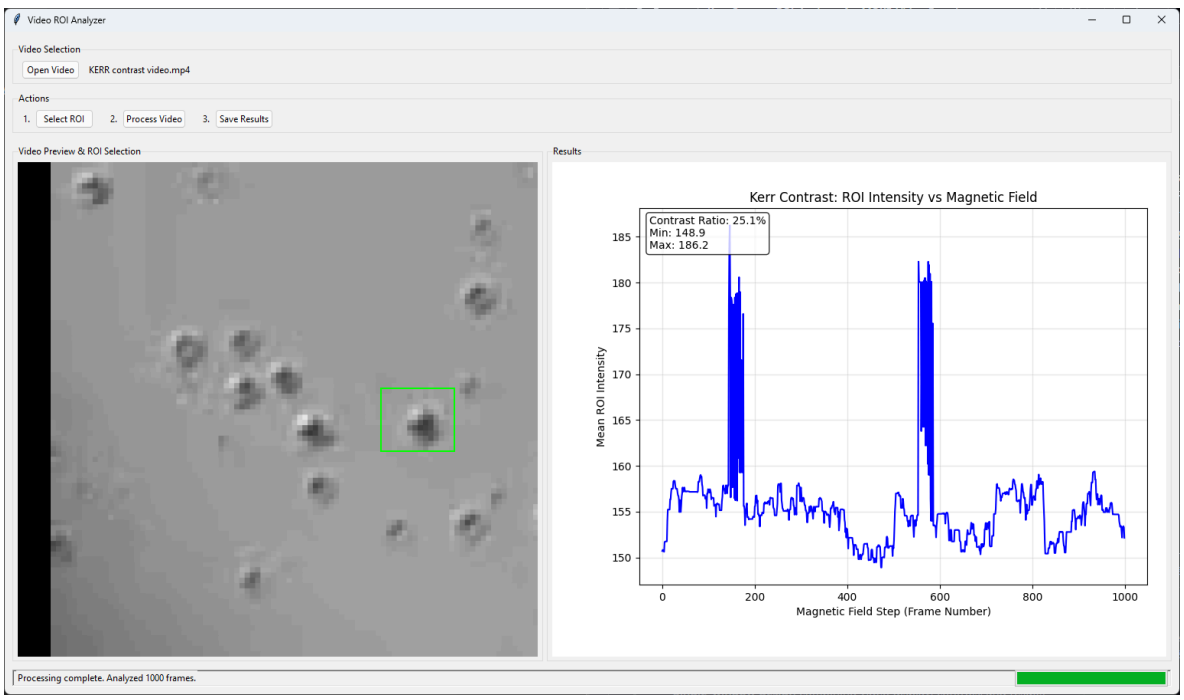
Past Projects Portfolio

Kerr Contrast ROI Analyzer for MOKE Video Experiments

During my internship at the National Research and Innovation Agency in Indonesia (BRIN), I built this GUI to analyze the Region of Interest (ROI) in the Magnetic-Optic Kerr Microscopy video to study the Kerr contrast during magnetic field sweeps. This application provides real-time video preview, interactive ROI selection, automatic frame processing, and data visualization for magneto-optical research applications. I built this with Python, OpenCV, numpy, pandas, matplotlib, and the tkinter module for the GUI.

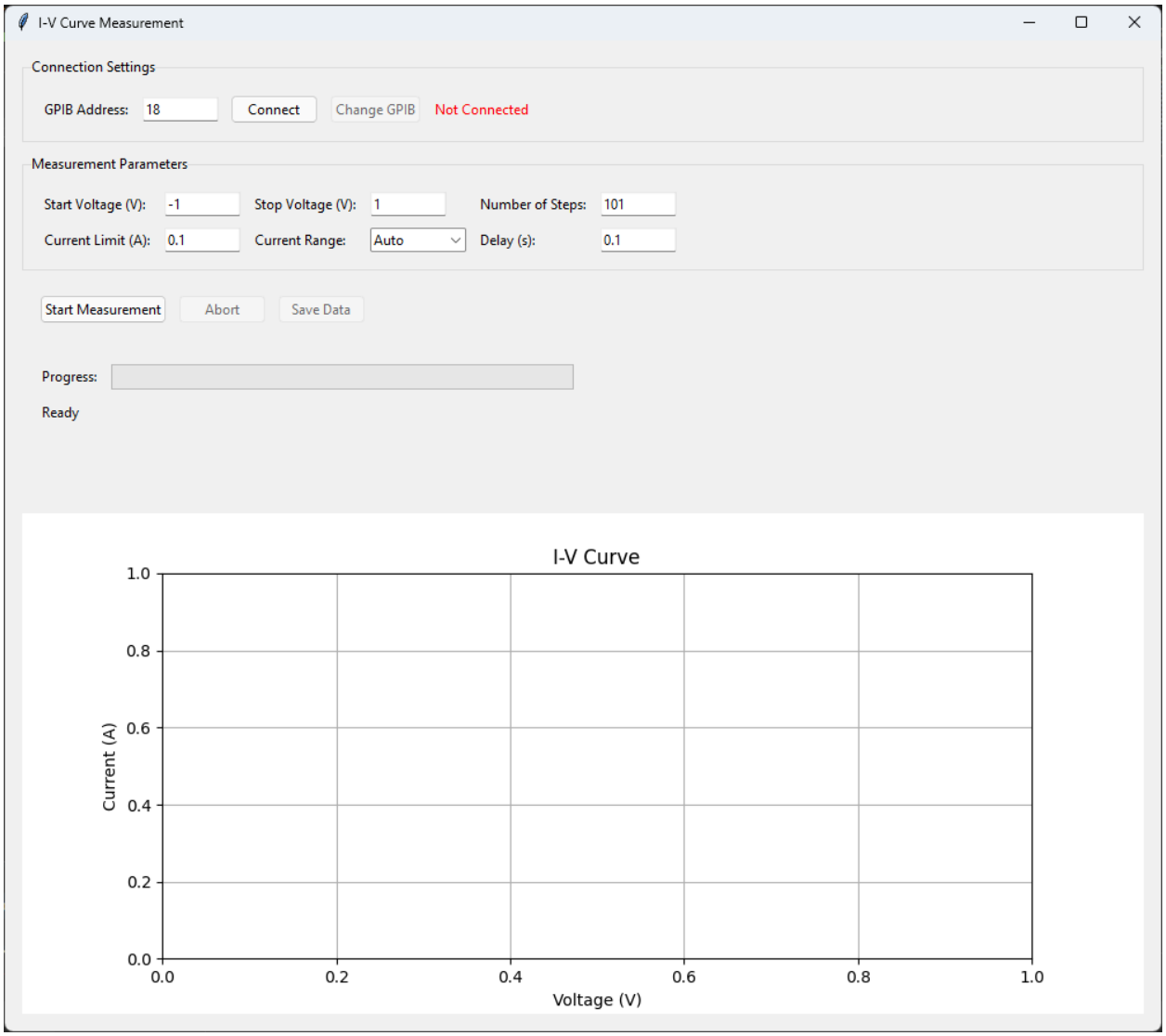


- Source Code: https://github.com/dietrichtarigan/Kerr-Contrast-ROI-Analyzer-for-MOKE-Video-Experiments

Keithley 2450 SMU I-V Curve Measurement Application

During my internship at the National Research and Innovation Agency in Indonesia (BRIN), I built a modular, fully configurable GUI application for I-V curve measurements

using the Keithley 2450 Source Measure Unit (SMU) through a GPIB interface. The application lets users perform voltage sweeps, measure currents, calculate resistances, and visualize I-V curves in real time. It was tested on a diode, and the results closely matched the SMU data, with the I-V curve updating smoothly and accurately.

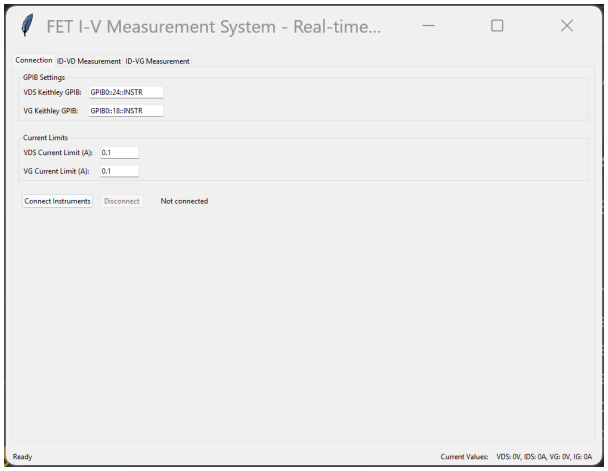


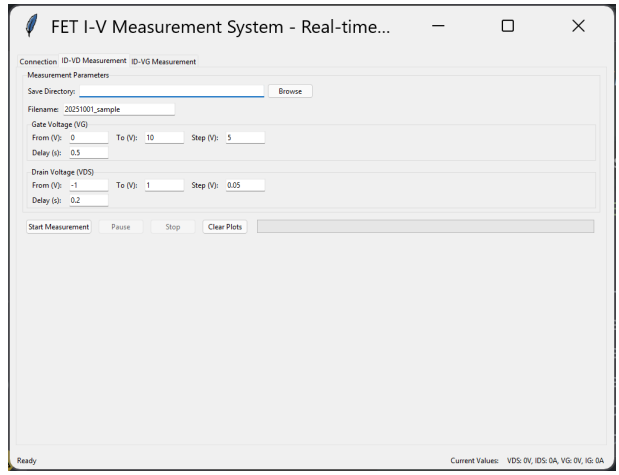
- Source Code: https://github.com/dietrichtarigan/Keithley-2450-I-V-Measurement-Program

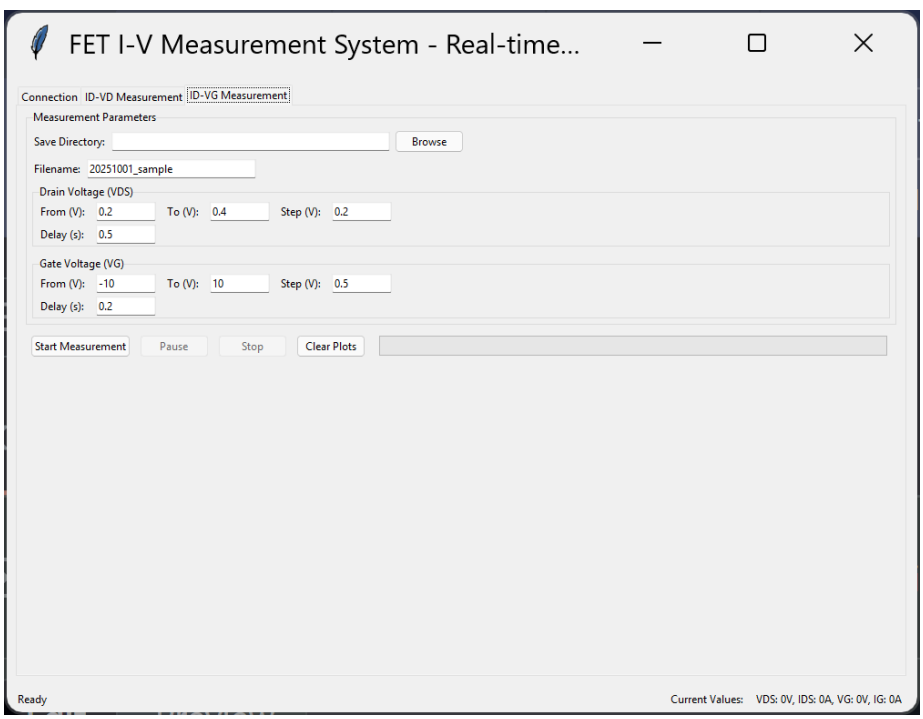
Keithley 2450 SMU FET Measurement Application

During my internship at the National Research and Innovation Agency in Indonesia (BRIN), I also built a multi-threaded GUI application using Tkinter to automate the electrical characterization of Field-Effect Transistors (FETs). The application uses the pyvisa library to communicate with Keithley 2450 SourceMeter instruments via GPIB,

running programmable ID-VD and ID-VG voltage sweeps in background threads to keep the interface responsive. As the sweeps run, the application logs voltage and current data to CSV files using Pandas and provides live, real-time visual feedback of the measurements via dynamic Matplotlib plots.







- Source Code: https://github.com/dietrichtarigan/FET-measurement